Joseph A. Smith

Environmental Systems & Intelligence Specialist | Infrastructure | Applied AI & Geospatial Systems

Norton Shores, MI | (231) 329-4858 | nvirogear1@gmail.com | josephsmith.info (in setup) — Target: Environmental technology, water infrastructure, remediation, technical operations

PROFILE

Environmental and infrastructure professional with hands-on wastewater, pump, mechanical and field operations experience, now building the technical systems that support them. Two and a half years as an environmental/wastewater field technician with Fleis & VandenBrink, assigned to the Zephyr remediation site in Muskegon: sampling, monitoring, lift station and pump inspection, mechanical and electrical diagnosis, and field documentation that holds up in agency review. Founder of NviroGear, building GIS-and-records systems that turn public environmental data into defensible reporting. Self-directed rather than degreed: learned these systems in the field and in the work.

CORE STRENGTHS

Wastewater & Pump Systems | Environmental Sampling & Compliance | GIS / Geospatial Workflows | Environmental Data Architecture | Applied AI & System Design | Mechanical / Electrical Troubleshooting | Groundwater & Contamination Intelligence | Field Reporting & Evidence Documentation | Industrial Equipment & Safety | Rapid Self-Directed Learning

EXPERIENCE

Founder / Environmental Intelligence & Systems Designer

NviroGear LLC — Michigan | Recent

- Built environmental intelligence workflows combining public environmental records, parcel information, GIS layers, groundwater and well data, and contamination evidence into site-level decision support.
- Designed the Alice land and environmental intelligence system: traceable evidence, canonical records, map layers, public/private controls, reusable components.
- Developed data ingestion and normalization workflows for county parcel records, environmental datasets, well records and regulatory information; JSON/GeoJSON, schemas, APIs, web applications, cloud deployment.
- Applied AI as a build and research partner to prototype software, investigate problems, and document systems.
- Explored groundwater-gradient, contamination-proximity and property-intelligence methods; translated complex records into understandable reports and interactive maps.

Environmental / Wastewater Field Technician

Fleis & VandenBrink — 2.5 years, assigned to the Zephyr remediation site, Muskegon

- Performed environmental and wastewater field work at the Zephyr remediation site under Fleis & VandenBrink, one of the larger sediment and groundwater remediation efforts in Muskegon County.
- Collected and documented environmental samples and monitoring data with chain-of-custody discipline; field notes that hold up to agency and client review.
- Inspected and checked lift stations, pumps and related infrastructure; identified mechanical and electrical faults and escalated or repaired within scope.
- Read project drawings and specifications; coordinated daily field operations with engineers and site staff.
- Kept field work inside permit and safety lines, applying HAZWOPER 40-hour and stormwater training on site.

Owner / Operator - Arborist & Tree Climbing Services

Self-employed — 5+ years

- Ran all aspects of a tree service business: estimating, scheduling, crew supervision, equipment maintenance, safety, customer communication.
- Climbed and rigged removals and prunings around structures and utilities; maintained saws, rigging, chippers and trucks.
- Operated and maintained diesel equipment and small-engine tools; managed preventive maintenance schedules.

Diesel Mechanic / Heavy Equipment Technician — Rebuilt semi diesel engines; diagnostics and repair on heavy equipment.

Lead Framer - Commercial Construction — Led framing crews on commercial projects including Walmart builds; blueprint reading, lifts, layout, safety.

Material Handler / Manufacturing Support — Wacker Neuson operations support: forklifts, cranes, SAP inventory and material flow.

CDL Driver — Regional tractor-trailer operations for Team Lindy Trucking.

INDEPENDENT RESEARCH

Independent research in acoustics, recording and frequency analysis: measuring how sound behaves in physical spaces, including overlap, separation, sensing and reproduction of complex signals. Field-tested methodology: document, measure, repeat. Kept as an active practice.

CERTIFICATIONS

OSHA / HAZWOPER 40-Hour — Completed
Industrial Stormwater Certification — Completed
Underground Storage Tank Inspection Certification — Completed
Respiratory Protection / PPE Certification — Completed
CDL — Eligible for renewal

TECHNICAL

Pumps and pumping systems • Diesel engines and heavy equipment • Electrical panels and control troubleshooting • Mechanical diagnostics • Environmental monitoring and sampling • Blueprint and specification reading • GIS concepts and map layers • Data normalization and schema design • APIs, JSON/GeoJSON, web applications • AI-assisted prototyping

EDUCATION

Self-directed technical education; no traditional software or engineering degree. Built the skill base in the field and through independent study; learns unfamiliar systems by working them.